

Phase 9 - Document 8

Frontend Implementation Handbook and Codex Development Blueprint

Purpose

Provide the complete frontend implementation blueprint for Codex and development teams.

Technology Stack

Next.js, React, TypeScript, TailwindCSS, Zustand, TanStack Query, i18next and Framer Motion.

Folder Structure

Components, Layouts, Modules, Services, Hooks, Store, Themes, Locales, Utilities and Assets.

Design Tokens

Arena Blue, Arena Purple, Success, Warning and Danger token system.

Typography

Inter as primary font and Space Grotesk as secondary font.

Theme System

Dark and Light theme support from day one.

Localization

All text must use translation keys and support multi-language architecture.

Component Philosophy

Reusable, localized, responsive, accessible and theme-aware components.

Layouts

App Layout, Authentication Layout and Admin Layout.

Global Components

Buttons, Cards, Tables, Modals, Forms and Navigation components.

Dashboard Components

Welcome Banner, Wallet Card, Trust Card, Featured Games and Season Cards.

Maze Arena Components

Maze Canvas, Opponent Preview, Lives Display, Difficulty Display and Replay Indicator.

PvP Components

Match Cards, Match Found Modal and Rival Components.

Wallet Components

Balance Card, Deposit Widget, Withdrawal Widget and Transaction History.

Tournament Components

Tournament Cards, Brackets and Championship Displays.

Replay Components

Replay Viewer, Replay Timeline and Route Heatmaps.

Admin Components

Fraud Dashboard, Treasury Dashboard and Investigation Panels.

API Integration

Dedicated service layer for authentication, wallet, maze, leaderboard, replay and tournaments.

State Management

Zustand for client state and TanStack Query for server state.

Security Rules

Frontend never decides balances, rewards, trust scores or treasury values.

Accessibility

WCAG AA compliance and keyboard navigation support.

Performance

Lazy loading, code splitting, image optimization and caching.

Mobile Support

Responsive desktop, tablet and mobile support across all modules.

Codex Build Order

Foundation, Dashboard, Maze Arena, Wallet, Tournaments, Replay Center and Admin Portal.

Success Criteria

Codex can build the entire frontend from this specification without redesigning the architecture.